

SUMMER INTERNSHIP PROGRAM

Artificial Intelligence & High-Performance Computing

Powered by NVIDIA H200 GPU Infrastructure

Program Duration	2 Weeks (14 Days)
Credit Hours	2 Credits
Total Contact Hours	30 Hours
Format	Lecture + Hands-on GPU Lab
Hardware	NVIDIA H200 Tensor Core GPU
Target Audience	Undergraduate / Graduate Students
Prerequisites	Basic Python Programming

Presidency School of AI and Advanced Computing, Presidency University
Summer 2026

Prepared

Recommended & Verified

Approved

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Pro Vice-Chancellor

1. Program Overview

This two-week intensive summer internship is designed to provide students with a rigorous, hands-on introduction to Artificial Intelligence, Machine Learning, and GPU-accelerated computing. Students will progress from foundational AI theory through applied deep learning, culminating in independent capstone projects executed on state-of-the-art NVIDIA H200 Tensor Core GPU hardware.

The program integrates structured lecture sessions, coding workshops, guided lab exercises, and a final project presentation — all within a 30-hour contact-hour framework that satisfies 2 academic credit requirements.

1.1 Learning Objectives

Upon successful completion of this internship, students will be able to:

- Understand and explain core AI and machine learning concepts including supervised, unsupervised, and reinforcement learning
- Design, train, and evaluate neural network models using PyTorch and CUDA-optimized libraries
- Leverage NVIDIA H200 GPU architecture for high-performance parallel computing workloads
- Apply transformer-based architectures and large language model (LLM) fine-tuning techniques
- Profile GPU utilization, identify bottlenecks, and optimize model training pipelines
- Communicate AI project findings through written reports and oral presentations

1.2 Who Should Attend

This program is open to students who meet the following criteria:

- Undergraduate students (Junior/Senior level) or Graduate students in CS, EE, Data Science, or related fields
- Basic Python programming experience (loops, functions, classes, NumPy/Pandas)
- Familiarity with linear algebra and basic statistics is recommended but not required
- Motivation to explore AI research and industry applications

1.3 NVIDIA H200 GPU Infrastructure

Students will have exclusive access to NVIDIA H200 Tensor Core GPUs throughout the program. The H200 represents NVIDIA's most advanced AI training and inference hardware, offering:

H200 Specification	Detail
GPU Memory	141 GB HBM3e
Memory Bandwidth	4.8 TB/s
FP8 Tensor TFLOPS	3,958 TFLOPS (FP8)
FP16 / BF16 TFLOPS	1,979 TFLOPS
Interconnect	NVLink 4 / NVSwitch
Compute Architecture	NVIDIA Hopper (GH100)
Transformer Engine	Yes — optimized for LLM workloads

2. Program Structure & Credit Hours

The program is structured across 10 instructional days over 2 weeks, totaling 30 contact hours. The breakdown below details how hours are allocated across lectures, labs, and assessment:

Activity Type	Hours / Week	Total Hours	% of Program
Lecture / Theory Sessions	3 hrs	6 hrs	20%
Guided Coding Workshops	3 hrs	6 hrs	20%
GPU Hands-on Lab Sessions	5 hrs	10 hrs	33%
Mentored Project Work	3 hrs	6 hrs	20%
Assessment & Presentations	1 hr	2 hrs	7%
TOTAL	15 hrs/week	30 hrs	100%

Credit Equivalency: 30 contact hours = 2 academic credit hours (following standard 15:1 hour-to-credit ratio)

3. Detailed 2-Week Schedule

Week 1 — AI Foundations & Deep Learning Essentials

Week 1 focuses on establishing strong theoretical foundations and getting students comfortable with GPU-accelerated Python environments.

Day	Topics & Activities	Hours	Format
Day 1	Program Orientation AI Landscape & Ethics Python & CUDA Environment Setup	3 hrs	Lecture + Setup
Day 2	Machine Learning Fundamentals: Regression, Classification, Overfitting Sklearn on GPU	3 hrs	Lecture + Lab
Day 3	Neural Networks from Scratch Backpropagation PyTorch Basics on H200	3 hrs	Lecture + Lab
Day 4	Convolutional Neural Networks (CNNs) Image Classification GPU Profiling with Nsight	3 hrs	Lecture+ Lab
Day 5	Hands on Google Antigravity	3 hrs	Lecture + Lab + Quiz

Week 2 — Advanced AI, LLMs & Capstone Project

Week 2 advances students into cutting-edge topics including large language models, generative AI, and GPU optimization — culminating in a capstone demonstration.

Day	Topics & Activities	Hours	Format
Day 6	Transformer Architecture Deep Dive Attention Mechanisms BERT & GPT Anatomy	3 hrs	Lecture + Lab
Day 7	LLM Fine-Tuning with LoRA & PEFT Hugging Face on H200 Prompt Engineering	3 hrs	Lecture
Day 8	Generative AI: Diffusion Models & GANs Stable Diffusion on H200 GPU	3 hrs	Lecture + Lab
Day 9	GPU Optimization: Mixed Precision, FlashAttention, Tensor Parallelism Capstone Work	3 hrs	Lab + Project
Day 10	Capstone Project Presentations Peer Review Certificates & Closing Ceremony	3 hrs	Presentations

4. Curriculum Content Modules

Module 1: Introduction to Artificial Intelligence

- History and evolution of AI from expert systems to deep learning
- Types of AI: Narrow AI, General AI, and AGI concepts
- Machine learning paradigms: Supervised, Unsupervised, Self-supervised, and Reinforcement Learning
- AI ethics, bias, fairness, and responsible AI deployment
- Overview of the AI hardware ecosystem: CPUs vs GPUs vs TPUs vs NPUs

Module 2: Machine Learning Fundamentals

- Linear and logistic regression — gradient descent from scratch
- Decision trees, random forests, and ensemble methods
- Support vector machines and kernel methods
- Model evaluation: train/validation/test splits, cross-validation, ROC-AUC
- Feature engineering, normalization, and dimensionality reduction (PCA, t-SNE)
- Hands-on: Titanic survival prediction accelerated via cuML on H200

Module 3: Deep Learning & Neural Networks

- Perceptron and multi-layer perceptron (MLP) architecture
- Activation functions: ReLU, GELU, Swish — mathematical intuition
- Backpropagation and automatic differentiation with PyTorch Autograd
- Optimizers: SGD, Adam, AdamW, and learning rate scheduling
- Regularization: Dropout, Batch Normalization, Weight Decay
- Convolutional Neural Networks (CNNs): filters, pooling, receptive fields
- Transfer learning with ResNet, EfficientNet, and ViT on ImageNet

Module 4: Transformers & Large Language Models

- Attention mechanism: Scaled Dot-Product, Multi-Head Attention
- Positional encoding and token embedding strategies
- Encoder-only (BERT), Decoder-only (GPT), Encoder-Decoder (T5) architectures
- Pre-training objectives: Masked Language Modeling, Causal Language Modeling
- Parameter-Efficient Fine-Tuning: LoRA, QLoRA, Adapters, Prefix Tuning
- Prompt engineering patterns: zero-shot, few-shot, chain-of-thought
- Hands-on: Fine-tune LLaMA 3 on custom dataset using 4-bit QLoRA on H200

Module 5: Generative AI

- Variational Autoencoders (VAEs) — latent space and reconstruction
- Generative Adversarial Networks (GANs) — generator vs discriminator training
- Denoising Diffusion Probabilistic Models (DDPM) and DDIM samplers
- Stable Diffusion architecture: VAE + U-Net + CLIP text encoder
- Hands-on: Text-to-image generation with Stable Diffusion XL on H200
- Reinforcement Learning from Human Feedback (RLHF) — conceptual overview

Module 6: GPU Computing & Optimization

- NVIDIA Hopper architecture deep dive: SM, Warp, Tensor Cores, HBM3e
- CUDA programming model: grids, blocks, threads, shared memory
- Mixed precision training: FP32, FP16, BF16, FP8 with automatic loss scaling
- FlashAttention v2 and v3 — memory-efficient attention on H200
- Tensor parallelism, pipeline parallelism, and data parallelism with DeepSpeed
- GPU profiling with NVIDIA Nsight Systems and Nsight Compute
- Throughput vs latency optimization for inference serving (TensorRT, vLLM)

5. Hands-on GPU Lab Sessions

All lab sessions are conducted directly on NVIDIA H200 GPU nodes. Students access dedicated lab environments via JupyterHub provisioned on the GPU cluster. Each lab includes a starter notebook, guided instructions, stretch challenges, and a submission checkpoint.

#	Lab Title	GPU Task	Duration	Learning Outcome
1	GPU Environment Setup & CUDA Hello World	CUDA Kernel	2 hrs	Configure H200 environment; write first CUDA kernel
2	Image Classification with CNN on H200	PyTorch CNN	2 hrs	Train ResNet-18 on CIFAR-10; achieve >90% accuracy
3	NLP Sentiment Analysis with BERT	Transformer	2 hrs	Fine-tune BERT on IMDb reviews using Hugging Face
4	LLM Fine-Tuning with QLoRA	LLaMA 3	3 hrs	Fine-tune LLaMA 3 8B with 4-bit QLoRA on custom data
5	Stable Diffusion Image Generation	Diffusion Model	2 hrs	Generate & customize images with SDXL pipeline

6	GPU Profiling & Optimization Sprint	Nsight	2 hrs	Profile model bottlenecks; apply FlashAttention; benchmark speedup
7	Capstone Project GPU Execution	Open Choice	3 hrs	Implement, train, and evaluate independent AI project on H200

5.1 Lab Environment & Tools

- JupyterHub on dedicated H200 GPU cluster — 1 GPU per student workload
- CUDA Toolkit 12.x, cuDNN 9.x, cuBLAS, cuSPARSE
- PyTorch 2.x with CUDA 12 support, torchvision, torchaudio
- Hugging Face Transformers, Datasets, PEFT, TRL, Accelerate
- NVIDIA Nsight Systems, Nsight Compute for GPU profiling
- vLLM for high-throughput LLM inference serving
- Weights & Biases (W&B) for experiment tracking and visualization
- Git / GitHub for version control and lab submission

6. Capstone Project

The capstone project (Days 9–10) allows students to apply everything learned in an independent AI project executed on the H200 GPU. Projects are scoped to be achievable within the program duration and must demonstrate meaningful GPU utilization.

6.1 Project Tracks (Student Choice)

Track A — Computer Vision

- Object detection with YOLO v9 on a custom dataset
- Medical image segmentation using U-Net or SAM (Segment Anything Model)
- Video understanding with TimeSformer or VideoMAE

Track B — Natural Language Processing

- Domain-specific LLM fine-tuning (legal, medical, code, scientific)
- Question answering system with RAG (Retrieval-Augmented Generation)
- Multilingual text classification and cross-lingual transfer

Track C — Generative AI

- Custom ControlNet implementation for conditional image generation
- Music generation using AudioLDM2 or MusicGen on H200
- AI avatar creation with DreamBooth personalization

Track D — GPU Systems & Optimization

- Benchmark suite comparing FP32, FP16, BF16, and FP8 training precision
- Implement and profile a custom CUDA kernel for matrix operations
- Distributed training pipeline with DeepSpeed ZeRO-3 across multiple GPUs

6.2 Deliverables

1. GitHub repository with documented, reproducible code
2. Technical report (4–6 pages) covering architecture, methodology, and results
3. 10-minute presentation with live demo on H200 GPU
4. Peer review of one other student's project

7. Assessment & Grading

Assessment Component	Weight	Due Date
Daily Lab Submissions (7 labs × 5 pts)	35%	End of each lab day
Week 1 Quiz (AI Fundamentals)	10%	Day 5
Capstone Project — Code & Repository	20%	Day 9 (11:59 PM)
Capstone Technical Report	15%	Day 10 (Morning)
Capstone Presentation & Demo	15%	Day 10 (Afternoon)
Peer Review & Participation	5%	Day 10
TOTAL	100%	

8. Program Policies & Requirements

8.1 Attendance

- Attendance at all 10 instructional days is mandatory for credit
- Students missing more than 1 day without prior approval will not receive credit
- Late arrivals exceeding 30 minutes count as an absence
- Remote participation is not available — all sessions are in-person GPU lab sessions

8.2 GPU Resource Usage Policy

- Each student is allocated a dedicated GPU instance during lab hours only
- GPU instances are automatically suspended outside of scheduled lab sessions
- Data must be saved to persistent storage before session end — temporary storage is wiped
- Prohibited: cryptocurrency mining, unauthorized model training, or sharing credentials
- All GPU usage is logged for academic and security purposes

8.3 Academic Integrity

- All code submissions must be the student's own work — AI-generated code must be declared
- Collaboration is encouraged during workshops; final lab submissions and capstone are individual
- Plagiarism or unauthorized AI-generated capstone content will result in program disqualification

8.4 Technical Requirements

- Personal laptop with modern browser (Chrome / Firefox) — GPU lab runs via JupyterHub
- GitHub account for lab submission
- Stable internet connection (minimum 10 Mbps)
- No local GPU required — all compute is provided on the H200 cluster

9. Program Benefits & Outcomes

What Students Receive

- 2 transferable academic credit hours upon successful completion
- Completion certificate from the AI & GPU Computing Summer Internship Program

Career Pathways

- Machine Learning Engineer — model development, training pipeline optimization
- AI Research Scientist — academic and industrial research roles
- GPU Systems Engineer — HPC, cloud infrastructure, AI accelerator design
- Data Scientist — applied ML for business intelligence and analytics
- LLM Application Developer — RAG systems, chatbot development, AI products

10. Program - Faculty & Support

Program

The program is led by faculty with active research in AI systems, GPU computing, and applied deep learning. Instructors hold PhD-level qualifications with industry experience at leading AI labs and technology companies.

Mentorship

- Daily office hours: 30 minutes post-session
- Async Q&A via dedicated Slack workspace
- 1:1 project mentoring sessions (30 min each, Days 8 and 9)
- Guest lectures from NVIDIA engineers and AI industry practitioners

Trainer Details (Internal)

SI No	Name	Designation
1	Mr. Vinayak Raju Kage	Assistant Professor
2	Mr. Gyanesh	Assistant Professor
3	Mr. Sai Kumar	Assistant Professor
4	Dr. Vignesh R	Assistant Professor
5	Ms. Nisha	Assistant Professor
6	Mr. Francis	Assistant Professor
7	Ms. Harika	Assistant Professor
8	Ms Varalakshmi T	Assistant Professor

Trainer Details (External)

SI No	Name	Designation
1	Dhanush P	CEO Director Spheronix Technology
2	Ms Shaik Aliya Firdous	Trainee and Manager
3	Vishnu Gupta Karapakula	Developer and Architect

11. Registration & Logistics

Cohort Size	60 students per batch
Application Deadline	4 weeks prior to program start date
Selection Criteria	Academic standing, programming quiz, motivation statement
Program Fee	Internal PU Students - Free; External Participants - Rs. 2000/-
Location	GPU Computing Lab (DGL04 / DGL05)
Contact	nvidia.coe@presidencyuniversity.in +91 8755980646, +918088003960, +919701732127
Website	www.presidencyuniversity.in

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